LINEAR AND ROBUST CONTROL SYSTEMS

ELEC 9731

9A - Matrix Fraction Description

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Slide 1

Topics

1 Introduction

2 Matrix Fraction Description

3 Greatest Common Divisors

References

Kailath ch 6

Notation

$D = D(s), N = N(s)$ are polynomial matrices i.e. matrices of polynomials.

$P(s)$ is always a matrix of rational transfer functions.

What is a Polynomial?

$3+2s$ is a polynomial in s.
$4+\frac{3}{s}$ is <u>not</u> a polynomial in s.
It is a polynomial in s^{-1}.
It is also called a <u>rational</u> function of s and then rewritten
$4+\frac{3}{s}=\frac{3+4s}{s}$.

When we say $a(s)$ is a polynomial, we mean it is a polynomial in s.

Similarly when we say $N(s)$ is a polynomial matrix, we mean it is a polynomial matrix in s.

MATRIX FRACTION DESCRIPTION (MFD)

• This provides a theory for getting minimal realizations when roots of $d(s)$ are not distinct.

Recall rational $P(s) = \frac{N(s)}{d(s)} = \frac{\text{PM}}{\text{polynomial}}$

•Introduce

$$\begin{aligned} D_R(s) &= d(s)I_m \\ \Rightarrow P(s)_{p\times m} &= N(s)_{p\times m} D_R^{-1}(s)_{m\times m} \end{aligned}$$

This is called a right MFD of rational $P(s)$

•Similarly set

$$\begin{aligned} D_L(s) &= d(s)I_p \\ \Rightarrow P(s)_{p\times m} &= D_L^{-1}(s)_{p\times p} N(s)_{p\times m} \end{aligned}$$

This is called a left MFD of rational $P(s)$

•Algebra of both left and right MFDs is much the same. So generally we only discuss right MFDs.

MFD DEGREE

$$\begin{aligned}\deg D_R(s) &= \deg |D_R(s)| \\ &= \deg d^m(s) = mr \\ &= \text{dimension of block controller SS form}\end{aligned}$$

Similarly

$$\begin{aligned}\deg D_L(s) &= \deg |D_L(s)| \\ &= \deg d^p(s) = pr \\ &= \text{dim of block observer SS form}\end{aligned}$$

•Claim

Given a right MFD we can construct a controllable SS form

Given a left MFD we can construct an observable SS form

• Minimal Order

We expect the minimal order SS form has dimension = minimum possible degree of any possible MFD.

But how to find these?

POLYNOMIAL MATRICES

•A polynomial matrix (PM) is just a matrix of polynomials.

If $D(s)$ is a PM which is square then it has full generic rank iff

$$detD(s) = |D(s)| \neq 0$$

•We can then consider right MFDs

$$P(s) = N(s)D^{-1}(s)$$

and similarly left MFDs

•Suppose there is a PM $W(s)$ with

$$\begin{aligned} detW(s) &\neq 0 \text{ and} \\ \underset{p\times m}{N(s)} &= \underset{p\times m}{\overline{N}(s)} \; \underset{m\times m}{W(s)} \quad , \underset{m\times m}{D(s)} = \underset{m\times m}{\overline{D}(s)} \; \underset{m\times m}{W(s)} \end{aligned}$$

Then

$$ND^{-1} = \overline{N}W(\overline{D}W)^{-1} = \overline{N}WW^{-1}\overline{D}^{-1} = \overline{N}\overline{D}^{-1}$$

so the MFD is unchanged.

• We call $W(s)$ a right divisor of $N(s), D(s)$

POLYNOMIAL MATRICES Continued

•But now

$$\begin{aligned} |D| &= \left|\overline{D}\right| |W| \\ \Rightarrow deg\,|D| &= deg\left|\overline{D}\right| + deg\,|W| \\ &\geq deg\left|\overline{D}\right| \end{aligned}$$

•So the degree of the MFD can be reduced by removing right divisors. We can thus hope to find a greatest common right divisor (gcrd).

Then once it is extracted the MFD degree will drop to a minimum value.

• Note that the resulting MFD will not be unique because we can have unimodular matrices.

UNIMODULAR MATRICES (UM)

- A UM matrix is a PM with a constant determinant.
- Example9.1

$$\begin{aligned} W(s) &= \begin{bmatrix} s-1 & 3s+1 \\ s & 3s+4 \end{bmatrix} \\ |W(s)| &= (s-1)(3s+4) - s(3s+1) \\ &= 3s^2 + s - 4 - 3s^2 - s = -4 \end{aligned}$$

- $W(s)$ is unimodular iff $W^{-1}(s)$ is also a PM.

Proof:If: $\quad W^{-1}(s) = \dfrac{adjW(s)}{\det W(s)} = \dfrac{PM}{const}$

Only if: If both $W(s), W^{-1}(s)$ are PMs then

$$\begin{aligned} |W(s)| &= a_+(s) = \text{polynomial} \\ \left|W^{-1}(s)\right| &= a_-(s) = \text{polynomial} \end{aligned}$$

and

$$\begin{aligned} a_+(s)a_-(s) &= 1 \\ \Rightarrow a_+(s) &= const, a_-(s) = const., \text{QED.} \end{aligned}$$

COPRIME PMs

- We say PMs $N(s)_{p\times m}, D(s)_{m\times m}$ are right coprime if they have only unimodular right divisors.

- We say the MFD $N(s)D^{-1}(s)$ is irreducible if the PMs $N(s), D(s)$ are right coprime.

- Irreducible MFDs are not unique.

FINDING A GCRD

- Given PMs $N(s)_{p\times m}, D(s)_{m\times m}$ form

$$\begin{array}{cc} \left[\begin{array}{c} D(s) \\ N(s) \end{array}\right] & \begin{array}{c} m \\ p \end{array} \\ m & \end{array}$$

And find elementary row operations (e.g. row3$\rightarrow 3 \times row3 + s * row4$) to zero out the bottom p rows with each column remaining having minimum degree

$$\begin{array}{cccccc} & \begin{array}{cc} m & p \end{array} & m & & & \\ \begin{array}{c} m \\ p \end{array} & \left(\begin{array}{cc} U_{11} & U_{12} \\ U_{21} & U_{22} \end{array}\right) & \left(\begin{array}{c} D \\ N \end{array}\right) & = & \left(\begin{array}{c} R \\ 0 \end{array}\right) & \begin{array}{c} m \\ p \end{array} \end{array}$$

Then the PM $R(s)$ is a GCRD of N, D

FINDING A GCRD Continued

Also $U = \begin{pmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{pmatrix}$ is unimodular.

- Row operations are unimodular

e.g. $row_1 \to row_1$

$row_2 \to 3 \times row_2 - s \times row_1$

is represented by the matrix operation

$$\begin{pmatrix} 1 & 0 \\ -s & 3 \end{pmatrix} \begin{pmatrix} r_1^T(s) \\ r_2^T(s) \end{pmatrix} = \begin{pmatrix} r_1^T(s) \\ 3r_2^T - sr_1^T(s) \end{pmatrix}$$

and clearly $\begin{pmatrix} 1 & 0 \\ -s & 3 \end{pmatrix}$ is unimodular.

NB. When operating on e.g. row 2 we can only multiply row 2 by a constant not by a polynomial.

GCRD PROOF

Let

$$U^{-1} = \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{pmatrix}$$

Since U is unimodular each V_{ij} is a PM, then

$$\begin{aligned} \begin{pmatrix} D \\ N \end{pmatrix} &= \begin{pmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{pmatrix} \begin{pmatrix} R \\ 0 \end{pmatrix} \\ \Rightarrow D &= V_{11}R, N = V_{21}R \end{aligned}$$

So PM, R is a right divisor. But also

$$\begin{aligned} R &= U_{11}D + U_{12}N \\ 0 &= U_{21}D + U_{22}N \end{aligned}$$

So if PM, R_1 is another right divisor with $D = D_1R_1, N = N_1R_1$ then, $R = (U_{11}D_1 + U_{12}N_1)R_1$.

So PM, R_1 is a right divisor of R

So PM, R is a gcrd.

Example 9.2

Find gcrd of

$$D(s) = \begin{pmatrix} s^2 & -1 \\ -s & s^2 \end{pmatrix}, N(s) = \begin{pmatrix} s & -s \\ 0 & 1 \end{pmatrix}$$

We have (NB deg $detD = 4$)

$$\begin{bmatrix} D \\ N \end{bmatrix} = \begin{bmatrix} s^2 & -1 \\ -s & s^2 \\ s & -s \\ 0 & 1 \end{bmatrix}$$

Strategy

With a 0 in column 1 we can zero out second column. Then work on 1st column. As in the table below:

Elementary Row Operation	Result	
$\mathrm{row}_1 \rightarrow r_1 + r_4$	s^2	0
$r_2 \rightarrow r_2 - s^2 r_4$	$-s$	0
$r_3 \rightarrow r_3 + s r_4$	s	0
$r_4 \rightarrow r_4$	0	1
$r_1 \longrightarrow r_1 + s r_2$	0	0
$r_2 \rightarrow r_2 + r_3$	0	0
$r_3 \rightarrow r_3$	s	0
$r_4 \rightarrow r_4$	0	1
interchange rows i.e.		
$r_1 \rightarrow r_1 + r_3$	s	0
$r_2 \rightarrow r_2 + r_4$	0	1
$r_3 \rightarrow r_3$	s	0
$r_4 \rightarrow r_4$	0	1
$r_1 \rightarrow r_1$	s	0
$r_2 \rightarrow r_2$	0	1
$r_3 \rightarrow r_3 - r_1$	0	0
$r_4 \rightarrow r_4 - r_2$	0	0

Slide 14

Example 9.2 Continued

NB. In the table, once a row is altered one must do row operations with the altered row.

Now check that the results are PMs

$$
\begin{aligned}
\overline{N} &= NR^{-1} \\
&= \begin{pmatrix} s & -s \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1/s & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -s \\ 0 & 1 \end{pmatrix} \\
\overline{D} &= DR^{-1} \\
&= \begin{pmatrix} s^2 & -1 \\ -s & s^2 \end{pmatrix} \begin{pmatrix} 1/s & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} s & -1 \\ -1 & s^2 \end{pmatrix}
\end{aligned}
$$

NB deg $det\overline{D} = 3$

Left MFD from right MFD

When PM, $D(s)$ is nonsingular the gcrd construction also gives an irreducible left MFD.

Theorem. In the gcrd construction

(i) U_{22} is non-singular

(ii) $ND^{-1} = -U_{22}^{-1}U_{21} =$ left MFD

(iii) U_{21}, U_{22} are left coprime i.e., U_{22}^{-1}, U_{21} is irreducible.

(iv) If N, D are coprime $\deg|D| = \deg|U_{22}|$

Left MFD from right MFD Continued

Proof

Already saw $D = V_{11}R$. Now D nonsingular $\Rightarrow$ both V_{11}, R nonsingular. Next, partitioned determinant identity gives

$$|U| = |U_{22}| \left| V_{11}^{-1} \right|$$

But U is nonsingular so we get (i), we also have

$$0 = U_{21}D + U_{22}N \Rightarrow \text{ (ii)}$$

If N, D are coprime then R is unimodular,
$\Rightarrow deg\,|D| = \deg |V_{11}| = \deg |U_{22}| \Rightarrow (iv)$